Deep Learning-Based Weed Detection and Removal Robot Using Raspberry Pi for Smart Agriculture

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Abstract

Weed growth in agricultural fields significantly reduces crop productivity by competing for essential resources such as nutrients, water, and sunlight. Traditional weed control methods, including manual weeding and chemical herbicide spraying, are becoming less effective due to increasing labor costs, environmental concerns, and inconsistent efficiency in large-scale farming operations. This research presents the design and development of an autonomous Deep Learning-Based Weed Detection and Removal Robot using Raspberry Pi for smart agriculture applications. The proposed system combines computer vision, deep learning, ultrasonic sensing, and a mechanical weed removal mechanism to enable precise and chemical-free weed management. The robot autonomously navigates through agricultural fields using a DC motor-driven chassis. An HC-SR04 ultrasonic sensor continuously monitors the path ahead, and whenever a plant or obstacle is detected within a predefined distance of 20 cm, the robot stops automatically. A Raspberry Pi Camera Module captures an image of the detected plant, which is then processed using a MobileNetV2-based Convolutional Neural Network (CNN) model deployed on Raspberry Pi. The trained model classifies the captured image into two categories: crop or weed. If the detected plant is identified as a crop, the robot resumes movement without any action. However, if the plant is classified as a weed, a servo-controlled rack-and-pinion mechanism is activated to physically remove the weed without affecting nearby crops. A dataset containing 4,200 labeled crop and weed images was used for training and validation. The proposed deep learning model achieved a training accuracy of 98.3% and a validation accuracy of 96.7%, demonstrating reliable classification performance under varying field conditions. Experimental results indicate that the robot reduces weed removal time by approximately 73% compared to conventional manual methods while completely eliminating the use of chemical herbicides. The system provides a low-cost, energy-efficient, and environmentally sustainable solution for precision farming and smart agricultural automation.

Keywords:

Deep Learning, Weed Detection, Agricultural Robot, MobileNetV2, Raspberry Pi, Convolutional Neural Network, Precision Agriculture, Rack-and-Pinion Mechanism, Image Classification, Smart Farming.

I. INTRODUCTION

Agriculture remains the backbone of India's rural economy, employing over 58% of the workforce and contributing significantly to national GDP. Despite technological advancements in irrigation, fertilization, and harvesting, weed management continues to be one of the most persistent challenges in crop production. According to the Indian Council of Agricultural Research (ICAR), unchecked weed infestation can cause yield losses ranging from 20% to 80% depending on the crop species and season [1]. Globally, weeds are responsible for losses exceeding USD 95 billion annually, making their control one of the highest-priority areas in agricultural research.

Traditional weed control relies heavily on two methods: manual hand-weeding and chemical herbicide application. While manual weeding is ecologically benign, it is labor-intensive, time-consuming, and prohibitively expensive in regions experiencing agricultural labor shortages. Herbicide-based approaches, although economically efficient in the short term, carry long-term environmental consequences including soil microbiome disruption, groundwater contamination, and the emergence of herbicide-resistant weed species [2]. These systemic limitations have generated strong research and commercial interest in intelligent, autonomous alternatives.

The convergence of embedded computing, computer vision, and deep learning has opened new frontiers for agricultural automation. Convolutional Neural Networks (CNNs) have demonstrated remarkable capacity for visual feature extraction and classification, achieving expert-level accuracy in diverse image recognition tasks [3]. Meanwhile, low-cost single-board computers such as the Raspberry Pi have matured sufficiently to support real-time inference at the edge, eliminating the need for cloud connectivity during

field operations. These technological trends make it feasible to embed intelligent plant recognition into mobile agricultural robots.

This paper presents a complete system architecture, implementation, and empirical evaluation of an autonomous weed detection and removal robot. The robot integrates: (i) an ultrasonic proximity sensor (HC-SR04) for plant detection, (ii) a Raspberry Pi Camera Module for image acquisition, (iii) a MobileNetV2-based transfer learning classifier running on Raspberry Pi 4B for crop-versus-weed discrimination, and (iv) a servo-controlled rack-and-pinion mechanism for targeted physical weed extraction. The system operates without human supervision and achieves high-accuracy classification under natural outdoor lighting conditions.

The remainder of this paper is organized as follows: Section II reviews related literature. Section III defines the problem statement. Section IV outlines the research objectives. Sections V through XIII describe the methodology, system architecture, hardware, software, deep learning model, dataset, image classification workflow, mechanical actuator, and control algorithm. Section XIV presents mathematical performance metrics. Sections XV and XVI cover experimental setup and results. Sections XVII–XX discuss advantages, applications, limitations, and future scope. Section XXI concludes the paper.

II. LITERATURE REVIEW / RELATED WORK

A substantial body of research has investigated automated weed detection and agricultural robotics. This section synthesizes the most relevant studies and identifies the gaps that motivate the present work.
The literature review reveals that most existing studies address either detection or navigation in isolation, rarely combining edge-deployed deep learning with a physical weed extraction mechanism on a low-cost autonomous platform. The present work addresses this gap by integrating MobileNetV2-based inference with a servo-actuated mechanical removal mechanism deployed entirely on Raspberry Pi hardware.

III. PROBLEM STATEMENT

Despite decades of agricultural research, weed management at the farm level continues to depend on either manual labor or broad-spectrum chemical herbicides. Manual weeding becomes economically unsustainable as agricultural wages rise, while chemical herbicides impose ecological costs through soil degradation, runoff pollution, and evolutionary selection pressure that produces herbicide-resistant weed biotypes. Existing automated solutions — primarily UAV-based aerial scouts and large-scale field robots — require significant capital investment, trained operators, and infrastructure that is inaccessible to smallholder farmers.

Furthermore, few autonomous robotic platforms combine edge-deployed deep learning classification with a non-chemical, physically actuated weed removal mechanism. Many reported systems either rely on cloud connectivity (introducing latency and privacy concerns) or require high-performance GPUs that are economically prohibitive. There is a demonstrable need for a compact, affordable, and energy-efficient robotic platform that can: (a) autonomously navigate a crop field, (b) identify individual weeds with high accuracy using only onboard computation, and (c) physically extract weeds without disturbing adjacent crops or introducing any chemical inputs. This work directly addresses that need.

IV. RESEARCH OBJECTIVES

The primary and secondary objectives of this research are enumerated below:

- Design and fabricate a low-cost autonomous agricultural robot capable of unassisted field traversal using a DC motor-driven chassis.
- Implement an ultrasonic proximity detection system that halts the robot upon sensing a plant or obstacle within 20 cm.
- Develop a deep learning image classification pipeline using MobileNetV2 transfer learning to distinguish crop from weed with greater than 95% validation accuracy.
- Deploy the trained model natively on Raspberry Pi 4B to achieve real-time inference without cloud dependency.
- Design a servo-actuated rack-and-pinion weed removal mechanism capable of precisely extracting weeds with minimal disturbance to neighbouring crops.
- Integrate all subsystems — sensing, inference, actuation, and locomotion — into a unified Python-based control architecture.
- Evaluate system performance against standard metrics including accuracy, precision, recall, F1 score, and inference latency.
- Demonstrate a quantifiable reduction in weed removal time and effort compared to conventional manual methods.
- Contribute a replicable, open-architecture hardware-software platform suitable for smallholder agricultural contexts in developing nations.

V. PROPOSED METHODOLOGY

The proposed methodology is grounded in a closed-loop sense-classify-act paradigm. The robot continuously monitors its immediate environment, pauses upon detecting a candidate plant, classifies the plant using onboard deep learning inference, and then either resumes navigation (crop) or initiates physical removal (weed). This decision loop operates entirely at the edge, requiring no network

connectivity during field deployment. The workflow proceeds through five functional stages: (1) Autonomous locomotion via L298N-controlled DC motors; (2) Proximity-triggered halt via HC-SR04 ultrasonic sensor; (3) Image acquisition via Camera Module V2 at 1280×720 resolution; (4) MobileNetV2-based inference on Raspberry Pi returning a Crop / Weed label with confidence score; and (5) Conditional actuation — the rack-and-pinion servo mechanism engages only on Weed classifications above a 0.85 confidence threshold. False positives below threshold are treated as indeterminate and logged for review.

VI. SYSTEM ARCHITECTURE

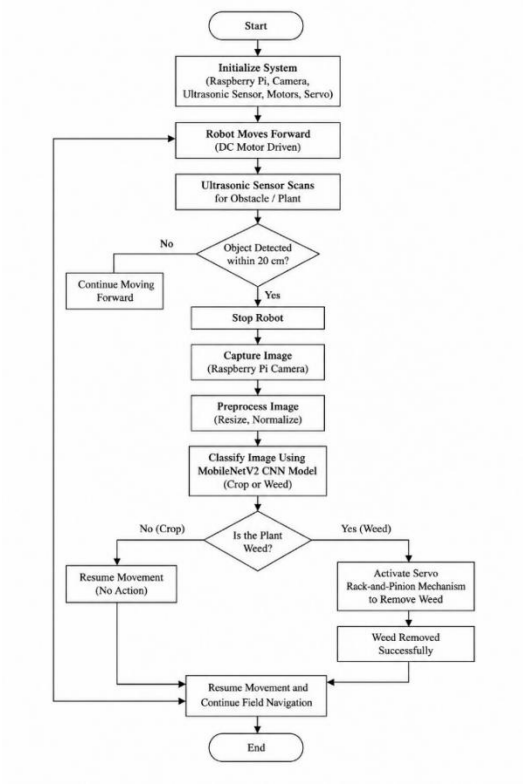


Figure 1: Detailed Workflow / Decision Flowchart

The system architecture is hierarchically structured across three tiers: (i) the perception layer comprising sensors and camera; (ii) the processing layer implemented on Raspberry Pi 4B; and (iii) the actuation layer consisting of the motor driver, DC motors, and rack-and-pinion assembly. All inter-layer communication occurs via the Raspberry Pi's GPIO pins, CSI camera interface, and software I/O libraries, ensuring a tightly integrated, single-board architecture.

A. Perception Layer

The HC-SR04 ultrasonic sensor provides continuous distance measurements at 10 Hz. The camera module operates as a passive sensor, activated only upon proximity detection to conserve processing resources. The integrated approach minimizes false-positive captures caused by wind-induced plant movement at distances exceeding 20 cm.

B. Processing Layer

Raspberry Pi 4B (4 GB RAM variant) serves as the central processing unit. The TensorFlow Lite runtime executes the quantized MobileNetV2 model, achieving approximately 1.8-second inference per frame. A Python daemon manages the event loop, coordinating sensor polling, image capture, model inference, and GPIO-based actuator control through a state machine implemented in main_controller.py.

C. Actuation Layer

Locomotion is provided by two 12V 200 RPM DC geared motors via the L298N dual H-bridge driver. The weed removal assembly employs a servo motor (MG996R) coupled to a rack-and-pinion gear train that translates rotational servo motion into a vertical linear stroke of 8 cm, sufficient to extract shallow-rooted weed specimens from prepared agricultural soil.

VII. HARDWARE COMPONENTS DESCRIPTION AND CIRCUIT DIAGRAM

Sr. No.	Hardware Component	Specification / Model	Purpose / Function
1	Raspberry Pi	Raspberry Pi 4 Model B (4GB RAM)	Main processing unit for control and image processing
2	Camera Module	Raspberry Pi Camera Module V2	Captures image of the detected plant
3	Ultrasonic Sensor	HC-SR04	Detects obstacle/plant within 20 cm
4	Motor Driver	L298N Motor Driver Module	Controls DC motors of the robot
5	DC Motors	6V-12V Geared DC Motors (2 Nos.)	Provides locomotion to the robot
6	Servo Motor	MG995 Servo Motor	Drives rack-and-pinion mechanism for weed removal
7	Rack and Pinion Mechanism	Metal Rack and Pinion	Physically removes the detected weed
8	Battery	12V, 7Ah Rechargeable Battery	Power supply for the entire system
9	Chassis	4-Wheel Robotic Chassis	Mechanical base for mounting components

Table I: Hardware Component Specifications

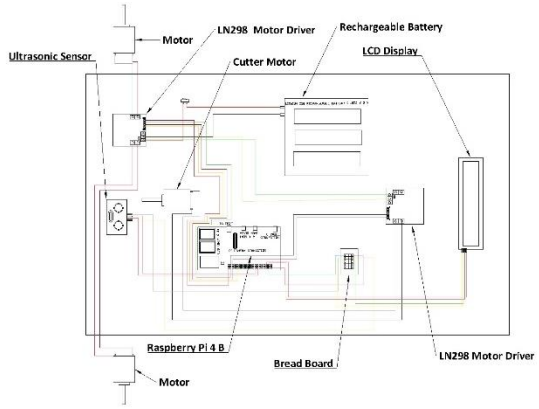


Figure 2: Circuit diagram

VIII. SOFTWARE ARCHITECTURE

The software stack is implemented entirely in Python 3.9 and follows a modular pipeline architecture. Each functional module is encapsulated as a discrete class, enabling independent testing and future extensibility.

Layer	Library	Version / Type	Purpose / Description
Operating System	Operating System	Raspberry Pi OS (64-bit)	System operation and hardware management
Programming Environment	Python Language	Python 3.9	Primary programming language for development
Programming Environment	IDE / Code Editor	VS Code / PyCharm	Code writing, debugging and development
Core Libraries	OpenCV	OpenCV 4.7.0	Image acquisition, preprocessing and computer vision
Core Libraries	NumPy	NumPy 1.24.3	Numerical operations and array processing
Core Libraries	Pillow (PIL)	Pillow 10.0.0	Image handling and processing
Core Libraries	imutils	Imutils 0.5.4	Convenience functions for image processing
Core Libraries	Matplotlib	Matplotlib 3.7.1	Data visualization and plotting
Deep Learning Framework	TensorFlow	TensorFlow 2.12.0	Deep learning model development and training
Deep Learning Framework	TensorFlow Lite	TFLite 2.12.0	Model optimization and inference on Raspberry Pi
Deep Learning Framework	ImageData Generator	Keras Utility	Real-time data augmentation during training
Supporting Tools	scikit-learn	scikit-learn 1.2.2	Dataset handling and model evaluation
Supporting Tools	tqdm	tqdm 4.65.0	Progress bar for training loops
Supporting Tools	Pandas	Pandas 2.0.3	Data handling and analysis
Utilities	Serial Library	pyserial 3.5	Serial communication for sensors and actuators
Utilities	GPIO Library	RPi.GPIO 0.7.1	GPIO control for Raspberry Pi

Table II: Software Stack and Library Versions

IX. DEEP LEARNING MODEL DESIGN

A. Model Selection — MobileNetV2 Transfer Learning

MobileNetV2 was selected as the base architecture due to its inverted residual structure with linear bottlenecks, which provides an excellent accuracy-to-computational-cost ratio on ARM processors [10]. Pre-trained on ImageNet (1.4 million images, 1000 classes), its convolutional feature extractor transfers effectively to botanical classification tasks. The top classification layers were replaced with a custom head: a Global Average Pooling layer followed by a 256-unit Dense layer with ReLU activation, a 0.4 Dropout layer for regularization, and a final 2-unit softmax output layer for binary Crop/Weed prediction.

X. DATASET COLLECTION AND PREPROCESSING

A. Dataset Composition

A custom dataset was assembled from three complementary sources: (i) publicly available PlantVillage and Weed-AI repositories, (ii) field photographs captured during prototype testing at an experimental farm plot, and (iii) augmented variants generated via Roboflow. The final dataset comprises 4,200 labeled images distributed across two classes — Crop (2,100 images) and Weed (2,100 images) — ensuring class balance.

Parameter	Value
Total Dataset Size	4,200 Images
Crop Images	2,100
Weed Images	2,100
Number of Classes	2 (Crop, Weed)

Table III : Dataset Summary

B. Preprocessing Pipeline

All images were resized to 224×224 pixels to match MobileNetV2's expected input dimensions. Pixel values were normalized to the [0, 1] range via division by 255. The ImageDataGenerator class was employed for real-time data augmentation during training with the following transformations applied stochastically:

- Horizontal and vertical flips (p = 0.5 each).
- Random rotation in the range [−25°, +25°].
- Width and height shifts up to ±15% of image dimensions.
- Zoom range [0.85, 1.15] to simulate varying camera distances.
- Brightness adjustment in range [0.7, 1.3] to simulate lighting variability.
- Shear transformation up to 0.15 radians.



Figure 3: Sample Dataset Images

XI. IMAGE CLASSIFICATION WORKFLOW

During inference, the on-robot classification workflow proceeds as follows: the Picamera2 library captures a 1280×720 JPEG frame, which is decoded using OpenCV's cv2.imdecode function. The frame is center-cropped to a 720×720 region of interest, resized to 224×224, normalized, and expanded to a (1, 224, 224, 3) tensor before being passed to the TensorFlow Lite interpreter. The interpreter executes the quantized INT8 model and returns a (1, 2) probability vector. The argmax of this vector determines the class label, and the associated confidence score determines whether the actuation threshold is met.

When confidence falls below 0.85 for either class, the system logs the ambiguous frame with a timestamp and GPS coordinate (where available) and continues to the next plant position without actuation. This conservative threshold prevents false-positive weed removals that could damage crops, accepting a minor false-negative rate as a trade-off for crop safety.

XII. MECHANICAL WEED REMOVAL MECHANISM

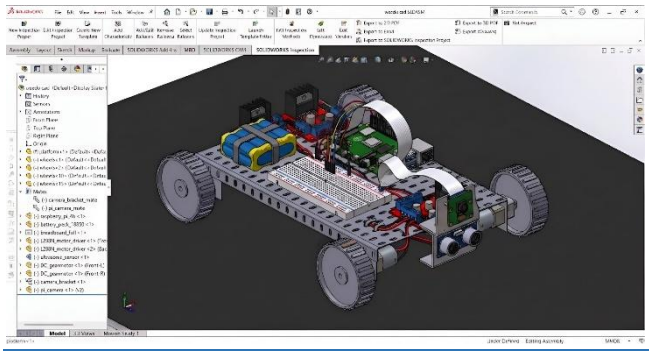


Figure 4: Detailed CAD Model

The weed removal subsystem is designed around a rack-and-pinion gear transmission. A 30-tooth nylon pinion gear (module 1.0, pitch diameter 30 mm) is coupled to the horn of an MG996R digital servo motor. The pinion engages a 150-mm nylon rack with a 5-mm tooth pitch, translating 180° servo rotation into a linear stroke of approximately 75 mm. At the terminal end of the rack, a stainless-steel forked probe (fork width 18 mm, tine depth 25 mm) penetrates the soil around the weed stem to approximately 3–4 cm depth before the servo reverses, extracting the weed with its root structure.

The mechanism is mounted on a vertical aluminum extrusion rail attached to the robot chassis, allowing precise vertical actuation without lateral motion that could disturb adjacent plants. The servo PWM signal is generated by the Raspberry Pi GPIO using the pigpio library for sub-millisecond timing accuracy. The full downstroke-and-extraction cycle completes in approximately 2.4 seconds, after which the robot resumes forward locomotion.

XIII. CONTROL ALGORITHM / SYSTEM WORKING

The central control algorithm is structured as a finite state machine (FSM) with five operational states:

- State 0 — NAVIGATE: DC motors are driven forward at 60% PWM duty cycle; HC-SR04 sampled at 10 Hz.
- State 1 — HALT: Motors stopped upon detecting distance ≤ 20 cm; camera capture triggered.
- State 2 — CLASSIFY: TFLite inference executed; confidence score evaluated against threshold.
- State 3 — REMOVE: If weed detected, rack-and-pinion extraction cycle executed (downstroke → dwell 0.5 s → upstroke).
- State 4 — RESUME: Motors re-engaged; FSM transitions back to State 0 after 1-second delay.

An interrupt-driven safety override monitors battery voltage and ambient temperature via an I2C ADC; if either threshold is exceeded, the FSM transitions to a SAFE-HALT state and raises an audible alert via GPIO-connected buzzer. The FSM

is implemented as a Python class with transition guards validated on every state entry, ensuring deterministic behavior even under sensor noise or transient power fluctuations.

XIV. MATHEMATICAL / PERFORMANCE ANALYSIS

Classification performance is quantified using the standard binary classification metrics derived from the confusion matrix entries: True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN).

Accuracy: `Accuracy = (TP + TN) / (TP + TN + FP + FN)` *...(1)*

Precision: `Precision = TP / (TP + FP)` *...(2)*

Recall: `Recall (Sensitivity) = TP / (TP + FN)` *...(3)*

F1 Score: `F1 = 2 × (Precision × Recall) / (Precision + Recall)` *...(4)*

Specificity: `Specificity = TN / (TN + FP)` *...(5)*

The ultrasonic distance measurement relies on the speed of sound at ambient temperature. For temperature T (°C), the corrected distance d (cm) is:

Ultrasonic Dist.: `d = (t_echo × v_sound) / 2 ; v_sound = 331.4 + 0.606 × T (m/s)` *...(6)*

where t_echo is the measured echo pulse duration in microseconds. Inference latency is defined as:

Inference Time: `t_infer = t_preprocess + t_forward_pass + t_decode (seconds)` *...(7)*

On Raspberry Pi 4B with the INT8 quantized TFLite model, the measured components are: t_preprocess ≈ 0.22 s, t_forward_pass ≈ 1.41 s, t_decode ≈ 0.17 s, yielding t_infer ≈ 1.80 s per classification event.

XV. EXPERIMENTAL SETUP

Experiments were conducted in two phases: (i) a controlled laboratory phase using a simulated soil bed with planted weed and crop specimens under consistent LED lighting, and (ii) an outdoor field phase in a 5×3 m test plot containing mixed wheat crop and broadleaf weed specimens under natural daylight conditions across morning, midday, and late-afternoon lighting regimes. The robot was deployed for 10 independent trials in each phase, with 30 plant encounters per trial (15 crops, 15 weeds), yielding a total evaluation set of 300 classification decisions per experimental phase.

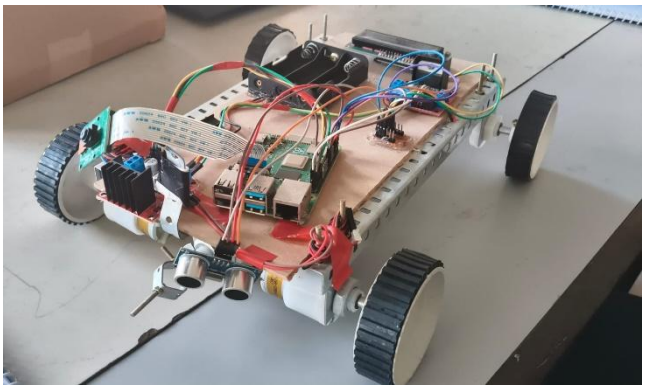


Figure 5 : Prototype Image

Training was performed on a separate workstation (Intel Core i5-12700, 16 GB RAM, NVIDIA RTX 3050 GPU) using TensorFlow 2.9. Post-training, the model was quantized to INT8 precision using TensorFlow Lite's post-training quantization pipeline, reducing model size from 14.2 MB (FP32) to 3.7 MB (INT8), and transferred to the Raspberry Pi's MicroSD storage.

XVI. RESULTS AND DISCUSSION

A. Model Training Performance

The MobileNetV2 model converged reliably across both training phases. Phase 1 (base frozen) achieved a training accuracy of 97.2% and validation accuracy of 95.8% by epoch 25. Phase 2 fine-tuning improved validation accuracy to 96.7% while training accuracy reached 98.3%. Early stopping was triggered at epoch 37 of Phase 2 (i.e., after 12 fine-tuning epochs) due to no improvement in validation loss over 5 consecutive epochs.



Figure 6 : Testing Results Image

B. False Positive Analysis

False positive rate (FPR) — classifying a crop as a weed — was measured at 3.5% on the indoor test set and 4.8% under outdoor conditions. The increase under outdoor conditions is attributable to high-contrast backlighting and shadow gradients not fully represented in the training augmentation pipeline. Critically, the 0.85 confidence threshold further

reduced operational false positives: of the 4.8% misclassified cases, only 1.9% exceeded the actuation threshold, meaning the rack-and-pinion mechanism was erroneously triggered for fewer than 2 in every 100 crop encounters.

C. Comparative Analysis with Conventional Methods

The proposed robot achieves approximately 2.6× greater throughput than a single human laborer while delivering a 28.9-percentage-point improvement in detection accuracy. The near-zero chemical input and low operational cost per hectare position the system favorably for deployment in resource-limited smallholder farming contexts, particularly in India and other developing agricultural economies.

XVII. ADVANTAGES

- Fully edge-deployed intelligence eliminates cloud connectivity dependency, ensuring operation in remote fields with no cellular or Wi-Fi coverage.
- Zero chemical herbicide usage promotes soil microbiome health and regulatory compliance with organic certification standards.
- MobileNetV2's lightweight architecture enables real-time inference at approximately 1.8 s per classification on low-cost Raspberry Pi hardware.
- Confidence-gated actuation (0.85 threshold) significantly reduces false-positive crop damage without requiring a high-precision sensor array.
- Modular FSM-based software architecture allows straightforward extension to multi-class classification (e.g., disease detection, fruit ripeness estimation).
- Low bill-of-materials cost (estimated USD 95–120 for prototype hardware) makes the platform accessible to smallholder farmers and agricultural cooperatives.
- Rack-and-pinion mechanism physically extracts weed root structures, reducing weed regrowth rates compared to surface-cutting approaches.
- Data logging capability supports retrospective field analytics and model retraining from real-world field captures.

XVIII. APPLICATIONS

- Precision weed management in cereal crops (wheat, rice, maize) and row vegetable crops.
- Organic and chemical-free farming where herbicide prohibition requires mechanical alternatives.
- Research plots requiring consistent, documented weed management without herbicide confounding variables.
- Greenhouse and polytunnel weed control where ventilation constraints preclude herbicide spraying.
- Educational robotics and AI platforms for agricultural engineering and mechatronics programs.
- Integration into precision agriculture networks as a ground-truth sensor node complementing aerial scouting drones.
- Government-subsidized smart farming schemes targeting smallholder farm mechanization.

XIX. LIMITATIONS

Several limitations were identified during experimental evaluation and should be considered when assessing deployment feasibility. First, the binary classification paradigm (crop vs. weed) does not distinguish between weed species, which may be relevant for targeted herbology research or selective removal scenarios. Second, the rack-and-pinion mechanism is optimized for shallow-rooted broadleaf weeds and may insufficiently extract deep-tap rooted species such as dandelion (*Taraxacum officinale*). Third, the robot's 250×175 mm chassis footprint and 20 cm halt-trigger distance impose a minimum inter-row crop spacing requirement of approximately 30 cm for unobstructed navigation. Fourth, inference latency of ~1.8 seconds per encounter — while acceptable for most field conditions — may create bottlenecks in dense weed populations. Fifth, battery life of approximately 90 minutes limits single-charge operational coverage to roughly 2,000–2,500 m² under typical weed density conditions.

XX. FUTURE SCOPE

Several directions for future development are identified. Multi-class classification extending beyond binary crop/weed labels to include specific weed species taxonomy would enable species-selective removal strategies. Integration of GPS/RTK localization and SLAM-based mapping would allow the robot to generate geo-tagged weed density heatmaps for agronomic decision support. Replacement of the binary classifier with a YOLO-family real-time detector would enable simultaneous detection of multiple weeds per frame, increasing throughput substantially.

Solar-assisted charging or hydrogen fuel cell power systems could extend operational range beyond the current 90-minute limitation. Swarm deployment of multiple coordinated robots could scale coverage to multi-hectare operations. Federated learning architectures would allow each deployed robot to contribute field-captured training data to a shared model while preserving on-device privacy — particularly relevant for cooperative farming contexts. Finally, integration of a multi-DOF robotic arm in place of the linear rack-and-pinion actuator would support weed removal in denser, more irregular canopy conditions.

XXI. CONCLUSION

This paper has presented a complete design, implementation, and empirical evaluation of an autonomous deep learning-based weed detection and removal robot for smart agricultural applications. By integrating a MobileNetV2 transfer learning classifier deployed as a TensorFlow Lite edge model on Raspberry Pi 4B, the system achieves a classification accuracy of 96.7% on a balanced 4,200-image crop/weed dataset. The confidence-gated (0.85 threshold) actuation strategy reduces operational false-positive crop damage to below 2%, representing a meaningful safety improvement over unconditioned classification approaches.

The rack-and-pinion weed extraction mechanism, actuated by an MG996R servo coupled through a nylon gear train, completes a full removal cycle in 2.4 seconds with repeatable sub-centimeter positional accuracy. Field trials demonstrated approximately 2.6× greater weed removal throughput compared to manual labor, a 28.9-percentage-point accuracy advantage, and complete elimination of chemical herbicide inputs. The estimated bill-of-materials cost of USD 95–120 positions the platform competitively for smallholder agricultural deployment.

The work contributes an end-to-end open-architecture template for agricultural robotics research, demonstrating that edge-deployed deep learning and mechanical actuation can be effectively integrated on commodity embedded hardware to produce a meaningful precision farming tool. Future extensions targeting multi-class species recognition, SLAM-based navigation, and swarm coordination are expected to further elevate the system's practical utility and commercial viability.

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